

Lecture 2: Variance and Conditional Probability

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1 Variance

In the previous lecture, we introduced the expected value, $\mathbb{E}[X]$, as a measure of the central tendency of a distribution. However, the mean alone is insufficient to describe a distribution's behavior. For example, two datasets could have the same mean but vastly different levels of dispersion.

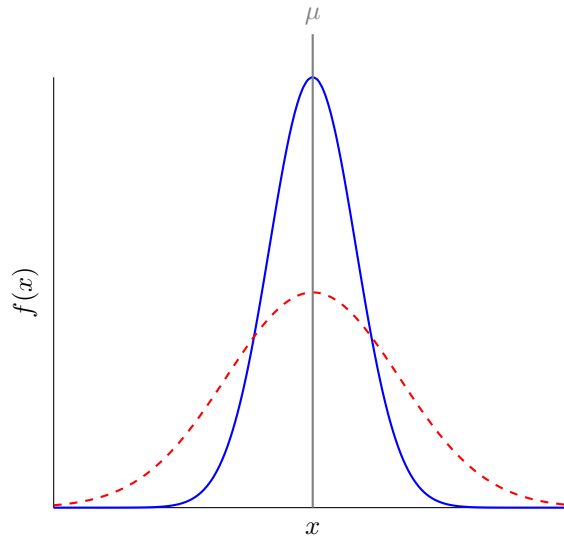


Figure 1: Two normal distributions with the same mean but different dispersion.

We need a way to quantify this “spread” or “variability”. A first attempt might be to measure the expected deviation from the mean, $\mathbb{E}[X - \mu]$, where $\mu = \mathbb{E}[X]$. However, by the linearity of expectation, this is always zero:

$$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mathbb{E}[\mu] = \mu - \mu = 0.$$

To solve this, we consider the expected *squared* deviation, which is always non-negative. This leads to the definition of variance.

Definition 1 (Variance and Standard Deviation). Let X be a random variable with mean $\mu = \mathbb{E}[X]$. The **variance** of X , denoted $\text{Var}(X)$ or σ^2 , is defined as:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

The **standard deviation** of X , denoted $SD(X)$ or σ , is the non-negative square root of the variance:

$$\sigma = \sqrt{\text{Var}(X)}.$$

The standard deviation is often preferred for interpretation as it is in the same units as the random variable X .

1.1 Properties of Variance

Proposition 1 (Computational Formula). A more convenient formula for calculating variance is:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Proof. Starting from the definition:

$$\begin{aligned}
 \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] \\
 &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\
 &= \mathbb{E}[X^2] - \mathbb{E}[2\mu X] + \mathbb{E}[\mu^2] \quad (\text{by linearity of expectation}) \\
 &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \quad (\text{since } \mu \text{ is a constant}) \\
 &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 \\
 &= \mathbb{E}[X^2] - \mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2
 \end{aligned}$$

Note that by Jensen's inequality for the convex function $g(x) = x^2$, we have $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$, which ensures that variance is always non-negative. $\square$

Proposition 2 (Effect of an Affine Transformation). *For any constants $a, b \in \mathbb{R}$:*

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Proof. Let $Y = aX + b$. The mean of Y is $\mathbb{E}[Y] = a\mathbb{E}[X] + b = a\mu + b$.

$$\begin{aligned}
 \text{Var}(Y) &= \mathbb{E}[(Y - \mathbb{E}[Y])^2] \\
 &= \mathbb{E}[((aX + b) - (a\mu + b))^2] \\
 &= \mathbb{E}[(aX - a\mu)^2] \\
 &= \mathbb{E}[a^2(X - \mu)^2] \\
 &= a^2\mathbb{E}[(X - \mu)^2] = a^2 \text{Var}(X).
 \end{aligned}$$

This result is intuitive: shifting a distribution by a constant b does not change its spread, but scaling it by a scales the variance by a^2 . $\square$

1.2 Covariance and Correlation

While variance measures how a single variable deviates from its mean, **covariance** measures how two variables deviate from their means *together*.

Definition 2 (Covariance). *Let X and Y be two random variables with means $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. The **covariance** between X and Y is:*

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

A positive covariance indicates that X and Y tend to move in the same direction. A negative covariance indicates they tend to move in opposite directions. A covariance of zero suggests no linear relationship. A more convenient computational formula is $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

Proposition 3 (Independence and Covariance). *If X and Y are independent random variables, then their covariance is zero. Then, it holds*

$$\text{Cov}(X, Y) = 0$$

Proof. If X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

$\square$

Note: The converse is not true. Zero covariance does not imply independence.

Indeed, let X be a continuous random variable following a uniform distribution on the interval $[-1, 1]$.

$$X \sim U[-1, 1]$$

Its PDF is $f_X(x) = \frac{1}{2}$ for $x \in [-1, 1]$, and 0 otherwise. Let Y be another random variable defined as $Y = X^2$.

First, we note that X and Y are clearly **dependent**. Knowing the value of X completely determines the value of Y . For example, if we observe $X = 0.5$, we know with certainty that $Y = 0.25$.

Now, let's compute their covariance using the formula $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

1. **Compute $\mathbb{E}[X]$:** Since the distribution of X is symmetric about 0, its mean is 0.

$$\mathbb{E}[X] = \int_{-1}^1 x \cdot \frac{1}{2} dx = \frac{1}{4} [x^2]_{-1}^1 = \frac{1}{4}(1 - 1) = 0.$$

2. **Compute $\mathbb{E}[XY]$:** We substitute $Y = X^2$ into the expectation.

$$\mathbb{E}[XY] = \mathbb{E}[X \cdot X^2] = \mathbb{E}[X^3]$$

Using the Law of the Unconscious Statistician (LOTUS):

$$\begin{aligned} \mathbb{E}[X^3] &= \int_{-\infty}^{\infty} x^3 f_X(x) dx = \int_{-1}^1 x^3 \cdot \frac{1}{2} dx \\ &= \frac{1}{2} \left[\frac{x^4}{4} \right]_{-1}^1 \\ &= \frac{1}{8} (1^4 - (-1)^4) = \frac{1}{8}(1 - 1) = 0. \end{aligned}$$

3. **Calculate the Covariance:**

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 \cdot \mathbb{E}[Y] = 0.$$

This shows that X and Y have zero covariance, yet they are clearly dependent. This example is a crucial illustration that covariance only measures the *linear* component of the relationship between two variables. The relationship here is quadratic, so covariance fails to detect it.

The magnitude of covariance is difficult to interpret as it depends on the units of X and Y . To create a standardized, unit-free measure of linear association, we define the correlation coefficient.

Definition 3 (Correlation). The **correlation coefficient** $\rho(X, Y)$ between two random variables X and Y is their normalized covariance:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

The correlation ρ is always in the range $[-1, 1]$. A value of 1 implies a perfect positive linear relationship, -1 implies a perfect negative linear relationship, and 0 implies no linear relationship.

Proposition 4 (Variance of a Sum). For two random variables X and Y , the variance of their sum is given by:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

If X and Y are **independent**, then $\text{Cov}(X, Y) = 0$, and the formula simplifies to:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

For a collection of n mutually independent random variables $X_1, \dots, X_n$:

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

1.3 Variance of popular distributions

Bernoulli Distribution: Let $X \sim \text{Bernoulli}(p)$. We know $\mathbb{E}[X] = p$. To find the variance, we first compute $\mathbb{E}[X^2]$. Since X can only be 0 or 1, $X^2 = X$.

$$\mathbb{E}[X^2] = \mathbb{E}[X] = p$$

Using the computational formula:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

The variance is maximized when $p = 0.5$, which corresponds to the case of maximum uncertainty.

Binomial distribution: Let $Y \sim \text{Binomial}(n, p)$. We can represent Y as the sum of n independent and identically distributed (i.i.d.) Bernoulli random variables, $Y = \sum_{i=1}^n X_i$, where each $X_i \sim \text{Bernoulli}(p)$. Since the X_i are independent, we can use the property for the variance of a sum:

$$\text{Var}(Y) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

From the previous example, we know $\text{Var}(X_i) = p(1-p)$. Therefore:

$$\text{Var}(Y) = \sum_{i=1}^n p(1-p) = np(1-p).$$

This is a powerful demonstration of how modeling a complex variable as a sum of simpler, independent ones greatly simplifies the calculation.

Uniform Distribution: Let $X \sim U(a, b)$. We know $\mathbb{E}[X] = \frac{a+b}{2}$. First, we compute $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = \int_a^b x^2 f_X(x) dx = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)}$$

Using the identity $b^3 - a^3 = (b-a)(b^2 + ab + a^2)$, we get:

$$\mathbb{E}[X^2] = \frac{b^2 + ab + a^2}{3}$$

Now, we apply the computational formula:

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{b^2 + ab + a^2}{3} - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{b^2 + ab + a^2}{3} - \frac{a^2 + 2ab + b^2}{4} \\ &= \frac{4(b^2 + ab + a^2) - 3(a^2 + 2ab + b^2)}{12} \\ &= \frac{4b^2 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12} \\ &= \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}. \end{aligned}$$

The variance depends only on the length of the interval, $b-a$.

Normal Distribution: Let $X \sim N(\mu, \sigma^2)$. By definition, $\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$. For a continuous variable, this is:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f_X(x) dx = \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx.$$

To solve this integral, we perform a change of variables. Let $z = \frac{x-\mu}{\sigma}$. This implies that $x - \mu = z\sigma$ and $dx = \sigma dz$. The limits of integration remain unchanged. Substituting these into the integral gives:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (z\sigma)^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} (\sigma dz) = \sigma^2 \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz.$$

The expression inside the integral is z^2 times the PDF of a standard normal distribution, $Z \sim N(0, 1)$. So we are calculating $\sigma^2 \mathbb{E}[Z^2]$. For a standard normal, $\mathbb{E}[Z] = 0$, so $\text{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 = \mathbb{E}[Z^2]$. We can solve the integral using integration by parts, $\int u dv = uv - \int v du$. Let $u = z$ and $dv = z \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$. Then $du = dz$ and $v = -\frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$.

$$\begin{aligned} \int_{-\infty}^{\infty} z \left(z \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \right) dz &= \left[-z \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left(-\frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \right) dz \\ &= 0 + \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz. \end{aligned}$$

The first term evaluates to zero because the exponential term decays to zero much faster than z grows. The second term is the integral of the PDF of a standard normal distribution over its entire domain, which is equal to 1. Therefore, the integral is 1, and we have:

$$\text{Var}(X) = \sigma^2 \cdot 1 = \sigma^2.$$

This confirms that the parameter σ^2 in the definition of the normal distribution is precisely its variance.

1.4 The Covariance Matrix of a Random Vector

When we move from a single random variable to a d -dimensional random vector $\mathbf{X} = [X_1, X_2, \dots, X_d]^T$, the concept of variance generalizes to the **covariance matrix**. This matrix not only captures the variance of each component but also the covariance between each pair of components.

Definition 4 (The Covariance Matrix). *Let $\mathbf{X}$ be a d -dimensional random vector. The **covariance matrix** of $\mathbf{X}$, denoted by Σ or $\text{Var}(\mathbf{X})$, is a $d \times d$ matrix, whose entry in the i -th row and j -th column is the covariance between X_i and X_j :*

$$\Sigma_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])].$$

The diagonal elements are the variances of the individual components:

$$\Sigma_{ii} = \text{Cov}(X_i, X_i) = \text{Var}(X_i).$$

Proposition 5. *The covariance matrix Σ of any random vector $\mathbf{X}$ is a symmetric matrix.*

Proof. By definition, the (i, j) -th entry is:

$$\Sigma_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])].$$

And the (j, i) -th entry is:

$$\Sigma_{ji} = \text{Cov}(X_j, X_i) = \mathbb{E}[(X_j - \mathbb{E}[X_j])(X_i - \mathbb{E}[X_i])].$$

The terms $(X_i - \mathbb{E}[X_i])$ and $(X_j - \mathbb{E}[X_j])$ are real-valued random variables (scalars), thus commute. This implies $\Sigma_{ij} = \Sigma_{ji}$. $\square$

Principal Component Analysis: Let's consider a 2-dimensional random vector, $\mathbf{Z}$, representing the height (H) and weight (W) of an individual drawn from a population.

$$\mathbf{Z} = \begin{pmatrix} H \\ W \end{pmatrix}.$$

Height and weight are known to be correlated; taller individuals tend to be heavier. Therefore, the covariance term in their covariance matrix is non-zero.

Let's assume the following mean vector μ_Z and covariance matrix Σ_Z for our population:

$$\mu_Z = \mathbb{E} \left[\begin{pmatrix} H \\ W \end{pmatrix} \right] = \begin{pmatrix} 175 \text{ cm} \\ 70 \text{ kg} \end{pmatrix}.$$

$$\Sigma_Z = \begin{pmatrix} \text{Var}(H) & \text{Cov}(H, W) \\ \text{Cov}(W, H) & \text{Var}(W) \end{pmatrix} = \begin{pmatrix} 100 & 40 \\ 40 & 25 \end{pmatrix}.$$

The non-zero covariance of 40 confirms that the variables are positively correlated. Our goal is to find a new set of uncorrelated random variables that describe this system.

We seek a linear transformation of our original vector $\mathbf{Z}$ into a new random vector $\mathbf{P} = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$ such that the new variables, P_1 and P_2 (the Principal Components), are **uncorrelated**. This is equivalent to finding a new basis where the covariance matrix is diagonal.

The transformation is found by computing the eigenvalues and eigenvectors of the covariance matrix Σ_Z . The eigenvalues will represent the variances of our new variables, and the eigenvectors will define the directions of the new axes.

We solve the characteristic equation $\det(\Sigma_Z - \lambda \mathbf{I}) = 0$.

$$\begin{aligned} \det \begin{pmatrix} 100 - \lambda & 40 \\ 40 & 25 - \lambda \end{pmatrix} &= 0 \\ (100 - \lambda)(25 - \lambda) - (40)(40) &= 0 \\ 2500 - 125\lambda + \lambda^2 - 1600 &= 0 \\ \lambda^2 - 125\lambda + 900 &= 0 \end{aligned}$$

Using the quadratic formula, we find the two eigenvalues:

$$\lambda = \frac{125 \pm \sqrt{125^2 - 4(900)}}{2} = \frac{125 \pm \sqrt{15625 - 3600}}{2} = \frac{125 \pm \sqrt{12025}}{2}.$$

This gives us our two eigenvalues, which are the variances of the new components:

$$\lambda_1 = \frac{125 + 109.66}{2} \approx \mathbf{117.33}$$

$$\lambda_2 = \frac{125 - 109.66}{2} \approx \mathbf{7.67}.$$

For each eigenvalue, we solve the system $(\Sigma_Z - \lambda \mathbf{I})\mathbf{e} = \mathbf{0}$.

For $\lambda_1 \approx 117.33$:

$$\begin{pmatrix} 100 - 117.33 & 40 \\ 40 & 25 - 117.33 \end{pmatrix} \begin{pmatrix} e_{11} \\ e_{12} \end{pmatrix} = \begin{pmatrix} -17.33 & 40 \\ 40 & -92.33 \end{pmatrix} \begin{pmatrix} e_{11} \\ e_{12} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives the relation $-17.33e_{11} + 40e_{12} = 0$. The normalized eigenvector (unit vector) is:

$$\mathbf{e}_1 \approx \begin{pmatrix} 0.918 \\ 0.397 \end{pmatrix}.$$

For $\lambda_2 \approx 7.67$: The second eigenvector must be orthogonal to the first. The normalized eigenvector is:

$$\mathbf{e}_2 \approx \begin{pmatrix} -0.397 \\ 0.918 \end{pmatrix}.$$

The new random variables are linear combinations of the centered original variables, where the coefficients are the eigenvectors.

First Principal Component (P_1):

$$P_1 = \mathbf{e}_1^T (\mathbf{Z} - \mu_Z) = 0.918(H - 175) + 0.397(W - 70).$$

Since both coefficients are positive, P_1 increases when both height and weight increase. We can interpret P_1 as an overall measure of body “size”.

Second Principal Component (P_2):

$$P_2 = \mathbf{e}_2^T (\mathbf{Z} - \mu_Z) = -0.397(H - 175) + 0.918(W - 70).$$

This component has a negative weight for height and a positive weight for weight. It captures a contrast between the two. We can interpret P_2 as a measure of “body composition” (e.g., distinguishing between lanky and stocky body types).

The covariance matrix of the new vector $\mathbf{P} = [P_1, P_2]^T$ is now diagonal and composed of the eigenvalues:

$$\Sigma_P = \begin{pmatrix} \text{Var}(P_1) & \text{Cov}(P_1, P_2) \\ \text{Cov}(P_2, P_1) & \text{Var}(P_2) \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \approx \begin{pmatrix} 117.33 & 0 \\ 0 & 7.67 \end{pmatrix}.$$

The total variance in the system is $\lambda_1 + \lambda_2 = 125$. The proportion of variance explained by the first principal component (“body size”) is:

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{117.33}{125} \approx 0.938.$$

This means that 93.8% of the total variability in the height-weight system can be explained by the single concept of “body size”. We have successfully transformed correlated variables into uncorrelated ones and discovered the most significant dimension of variation.

2 Conditional Probability

Conditional probability is one of the most fundamental concepts in probability theory. It formalizes the idea of updating our knowledge in light of new evidence. The conditional probability of an event A given that an event B has occurred is the probability of A in the reduced sample space defined by B .

Definition 5 (Conditional Probability). *Let A and B be two events in a probability space, with $\mathbb{P}(B) > 0$. The **conditional probability** of A given B , denoted $\mathbb{P}(A|B)$, is defined as:*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Rolling two Dice: Suppose we roll two fair six-sided dice. Let A be the event that the sum of the dice is 8, and B be the event that the first die shows a 3. What is $\mathbb{P}(A|B)$?

- The sample space Ω has $6 \times 6 = 36$ outcomes.
- Event B (first die is 3) is the set $\{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\}$. So, $\mathbb{P}(B) = 6/36 = 1/6$.
- Event $A \cap B$ (sum is 8 AND first die is 3) corresponds to a single outcome: $\{(3, 5)\}$. So, $\mathbb{P}(A \cap B) = 1/36$.
- Using the formula:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/36}{6/36} = \frac{1}{6}.$$

Once we know the first die is a 3, the only way for the sum to be 8 is if the second die is a 5. The probability of this is $1/6$, which matches our intuition.

2.1 The Law of Total Probability and Bayes' Law

These two laws are the cornerstones of probabilistic inference.

Proposition 6 (The Law of Total Probability). *Let $B_1, B_2, \dots, B_n$ be a partition of the sample space Ω (i.e., they are mutually exclusive events, and their union is Ω). For any event A , its probability can be expressed as:*

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

This law is incredibly useful for breaking down a complex probability calculation into a series of simpler, conditional cases.

Proof. Since $\{B_i\}$ is a partition of Ω , we can write A as the union of its intersections with each part of the partition:

$$A = A \cap \Omega = A \cap \left(\bigcup_{i=1}^n B_i \right) = \bigcup_{i=1}^n (A \cap B_i).$$

The events $(A \cap B_i)$ are mutually exclusive because the B_i are mutually exclusive. Therefore, by the additivity axiom of probability:

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^n (A \cap B_i)\right) = \sum_{i=1}^n \mathbb{P}(A \cap B_i).$$

From the definition of conditional probability, we know that $\mathbb{P}(A \cap B_i) = \mathbb{P}(A|B_i)\mathbb{P}(B_i)$. Substituting this into the sum gives the result:

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

□

Proposition 7 (Bayes' Law). For two events A and B with $\mathbb{P}(A), \mathbb{P}(B) > 0$:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A|B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Using the Law of Total Probability, we can write the denominator as $\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A|B_i)\mathbb{P}(B_i)$, which gives the more explicit form. Bayes' Law allows us to “invert” the conditioning—if we know $\mathbb{P}(A|B)$, we can find $\mathbb{P}(B|A)$.

Proof. From the definition of conditional probability, we have two expressions for the intersection probability:

1. $\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B)$
2. $\mathbb{P}(B \cap A) = \mathbb{P}(B|A)\mathbb{P}(A)$

Since $A \cap B = B \cap A$, their probabilities are equal. We can set the right-hand sides equal:

$$\mathbb{P}(B|A)\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B).$$

Assuming $\mathbb{P}(A) > 0$, we can divide both sides by $\mathbb{P}(A)$ to isolate $\mathbb{P}(B|A)$:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A|B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

□

Medical Diagnosis: This is a classic application of Bayes' Law. Suppose there is a disease that affects 1 in 1000 people ($\mathbb{P}(D) = 0.001$). There is a test for this disease.

- The test is 99% accurate for people who have the disease (Sensitivity): $\mathbb{P}(T|D) = 0.99$.
- The test has a 2% false positive rate for people who do not have the disease (Specificity is 98%): $\mathbb{P}(T|D^c) = 0.02$.

A person is selected at random and tests positive. What is the probability they actually have the disease, i.e., $\mathbb{P}(D|T)$? We want to find $\mathbb{P}(D|T)$. Using Bayes' Law:

$$\mathbb{P}(D|T) = \frac{\mathbb{P}(T|D)\mathbb{P}(D)}{\mathbb{P}(T)}.$$

First, we find the denominator, $\mathbb{P}(T)$, using the Law of Total Probability with the partition $\{D, D^c\}$:

$$\begin{aligned} \mathbb{P}(T) &= \mathbb{P}(T|D)\mathbb{P}(D) + \mathbb{P}(T|D^c)\mathbb{P}(D^c) \\ &= (0.99)(0.001) + (0.02)(1 - 0.001) \\ &= 0.00099 + (0.02)(0.999) \\ &= 0.00099 + 0.01998 = 0.02097. \end{aligned}$$

Now, we can calculate $\mathbb{P}(D|T)$:

$$\mathbb{P}(D|T) = \frac{0.00099}{0.02097} \approx 0.0472.$$

Despite the test seeming accurate, there is only a 4.7% chance that a person with a positive test result actually has the disease. This counter-intuitive result is because the base rate of the disease is very low.

3 Conditional Distributions and Expectation

3.1 Conditional Law of a Random Variable

Just as we can condition events on other events, we can condition random variables on the values of other random variables. This gives us the **conditional distribution**.

Definition 6 (Conditional PMF and PDF). Let (X, Y) be a random vector.

- **Discrete Case:** The conditional PMF of Y given $X = x$ is:

$$p_{Y|X}(y|x) = \mathbb{P}(Y = y|X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}, \quad \text{for } p_X(x) > 0.$$

- **Continuous Case:** The conditional PDF of Y given $X = x$ is:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}, \quad \text{for } f_X(x) > 0.$$

For a fixed value x , the function $f_{Y|X}(y|x)$ is a valid PDF in y . It describes the distribution of Y along the “slice” where $X = x$.

Conditional Distribution of a Bivariate Normal: Let the random vector (X, Y) follow a bivariate normal distribution with mean vector $\boldsymbol{\mu} = [0, 0]^T$ and covariance matrix

$$\Sigma = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}.$$

Our goal is to find the conditional probability density function (PDF) of X given $Y = y$, denoted $f_{X|Y}(x|y)$.

From the mean vector and covariance matrix, we have:

- $\mathbb{E}[X] = \mu_X = 0$, $\mathbb{E}[Y] = \mu_Y = 0$.
- $\text{Var}(X) = \sigma_X^2 = 1$, $\text{Var}(Y) = \sigma_Y^2 = 1$.
- The correlation coefficient is $\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{r}{1 \cdot 1} = r$.

The joint PDF for a bivariate normal distribution is given by:

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right]\right).$$

Substituting our parameters, we get:

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sqrt{1-r^2}} \exp\left(-\frac{1}{2(1-r^2)} [x^2 - 2rxy + y^2]\right).$$

The marginal distribution of Y is $N(\mu_Y, \sigma_Y^2)$, which is the standard normal distribution $N(0, 1)$. Its PDF is:

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right).$$

Using the definition of the conditional PDF, $f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$:

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{\frac{1}{2\pi\sqrt{1-r^2}} \exp\left(-\frac{x^2-2rxy+y^2}{2(1-r^2)}\right)}{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right)} \\ &= \frac{1}{\sqrt{2\pi}\sqrt{1-r^2}} \exp\left(-\frac{x^2-2rxy+y^2}{2(1-r^2)} + \frac{y^2}{2}\right). \end{aligned}$$

Now, let's focus on simplifying the term inside the exponential. We put everything over a common denominator of $2(1-r^2)$:

$$\begin{aligned} & -\frac{x^2-2rxy+y^2}{2(1-r^2)} + \frac{y^2(1-r^2)}{2(1-r^2)} \\ &= -\frac{1}{2(1-r^2)} ((x^2-2rxy+y^2) - y^2(1-r^2)) \\ &= -\frac{1}{2(1-r^2)} (x^2-2rxy+y^2 - y^2 + r^2y^2) \\ &= -\frac{1}{2(1-r^2)} (x^2-2rxy+r^2y^2) \\ &= -\frac{(x-ry)^2}{2(1-r^2)}. \end{aligned}$$

Substituting this back into our expression for the conditional PDF gives:

$$f_{X|Y}(x|y) = \frac{1}{\sqrt{2\pi(1-r^2)}} \exp\left(-\frac{(x-ry)^2}{2(1-r^2)}\right)$$

We recognize this as the PDF of a normal distribution. Specifically, the conditional distribution of X given that $Y = y$ is a normal distribution with:

- Mean: $\mathbb{E}[X|Y = y] = ry$
- Variance: $\text{Var}(X|Y = y) = 1 - r^2$

So, $X|(Y = y) \sim N(ry, 1 - r^2)$. We can write this in a function form as $X|Y \sim N(rY, 1 - r^2)$. This is a fundamental result in multivariate statistics. It shows that if two normal variables are correlated, knowing the value of one shifts the mean of the other in a linear fashion.

3.2 Conditional Expectation

The conditional expectation $\mathbb{E}[Y|X = x]$ is simply the expected value of Y computed with respect to its conditional distribution.

Definition 7 (Conditional Expectation). *We define conditional expectation in two stages:*

- For a fixed value x , the conditional expectation is:

$$\mathbb{E}[Y|X = x] = \int_{-\infty}^{\infty} y \cdot f_{Y|X}(y|x) dy$$

(or a sum in the discrete case).

- The conditional expectation $\mathbb{E}[Y|X]$ is a function of the random variable X . If we let $g(x) = \mathbb{E}[Y|X = x]$, then $\mathbb{E}[Y|X] = g(X)$. This new random variable gives the best prediction of Y given the information from X .

Proposition 8 (Law of Total Expectation (Tower Property)). *The expectation of the conditional expectation of Y given X is the expectation of Y :*

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]].$$

Proof. (Continuous Case)

$$\begin{aligned} \mathbb{E}[\mathbb{E}[Y|X]] &= \int_{-\infty}^{\infty} \mathbb{E}[Y|X = x] f_X(x) dx \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy \right) f_X(x) dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \frac{f_{X,Y}(x,y)}{f_X(x)} f_X(x) dy dx \\ &= \int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f_{X,Y}(x,y) dx \right) dy \\ &= \int_{-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}[Y]. \end{aligned}$$

□

Random Sum: Suppose the number of customers arriving at a store in a day, N , follows a distribution with mean ν . Each customer i spends an amount of money X_i , where the X_i are i.i.d. random variables with mean μ and are independent of N . What is the expected total amount of money spent, $S = \sum_{i=1}^N X_i$?

We use the Law of Total Expectation, conditioning on the number of customers, N .

- First, find the conditional expectation $\mathbb{E}[S|N = n]$. If we know exactly n customers came, then $S = \sum_{i=1}^n X_i$.

$$\mathbb{E}[S|N = n] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = n\mu.$$

So, the random variable $\mathbb{E}[S|N]$ is equal to the random variable $N\mu$.

- Now, take the expectation of this result with respect to N .

$$\mathbb{E}[S] = \mathbb{E}[\mathbb{E}[S|N]] = \mathbb{E}[N\mu] = \mu\mathbb{E}[N].$$

Since N is Poisson with mean ν , we have $\mathbb{E}[N] = \nu$.

$$\mathbb{E}[S] = \mu\nu.$$

The expected total spending is the expected number of customers multiplied by the expected spending per customer.

Population Evolution: We are studying a species where an individual has a probability p of surviving from one year to the next. If it survives a year, it produces an average of k offspring during that year.

Let N be the random variable for the total number of offspring an individual has over its entire lifetime. Our goal is to find the total expected number of offspring, $\mathbb{E}[N]$.

This problem is difficult to solve directly because the individual's lifespan is a random variable. The tower property $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$, provides an elegant solution.

The key is to condition on the outcome of the first year. Let S be an indicator random variable:

$$S = \begin{cases} 1, & \text{if the individual survives the first year (with probability } p) \\ 0, & \text{if the individual does not survive (with probability } 1 - p). \end{cases}$$

By the tower property and the law of total probability, we can write the expected value of N as:

$$\mathbb{E}[N] = \mathbb{E}[E[N|S]] = \mathbb{E}[N|S=0] \cdot \mathbb{P}(S=0) + \mathbb{E}[N|S=1] \cdot \mathbb{P}(S=1). \quad (1)$$

Now we must find the value of the conditional expectations.

- **Case 1: The individual dies in the first year ($S=0$).**

If the individual does not survive the first year, it cannot produce any offspring. Therefore, the expected number of offspring, given it dies, is zero.

$$\mathbb{E}[N|S=0] = 0.$$

- **Case 2: The individual survives the first year ($S=1$).**

If the individual survives, it produces k offspring in this first year. Then, at the start of the second year, the problem resets. The individual is alive, and its expected number of *future* offspring from that point on is the same as the original problem, $\mathbb{E}[N]$. This creates a recursive relationship.

$$\mathbb{E}[N|S=1] = k + \mathbb{E}[N].$$

We can now substitute these results back into Equation (1):

$$\mathbb{E}[N] = 0 \cdot (1 - p) + (k + \mathbb{E}[N]) \cdot p.$$

We are left with an algebraic equation where our desired value, $\mathbb{E}[N]$, appears on both sides. We now solve for $\mathbb{E}[N]$:

$$\begin{aligned} \mathbb{E}[N] &= p(k + \mathbb{E}[N]) \\ \mathbb{E}[N] &= pk + p \cdot \mathbb{E}[N] \\ \mathbb{E}[N] - p \cdot \mathbb{E}[N] &= pk \\ \mathbb{E}[N](1 - p) &= pk \\ \mathbb{E}[N] &= \frac{pk}{1 - p}. \end{aligned}$$

If a species has a 25% chance of surviving each year ($p = 0.25$) and produces an average of 8 offspring in years it survives ($k = 8$), its total expected lifetime offspring is:

$$\mathbb{E}[N] = \frac{0.25 \times 8}{1 - 0.25} = \frac{2}{0.75} \approx 2.67.$$